For Examiner's Use

3 **(a)** State what is meant by *internal energy* of a gas.

……

……

…… [1]

(b) 1.2×10^{-5} mol of ideal monoatomic gas is held in a container which allows heat to be freely exchanged between the gas, a regenerator and the surroundings. The regenerator acts as a type of heat storage that allows the gas to transfer heat to and from the regenerator as the gas cycles through the 4 states. The gas is made to undergo the following thermodynamic cycle as shown in Fig. 3.1.

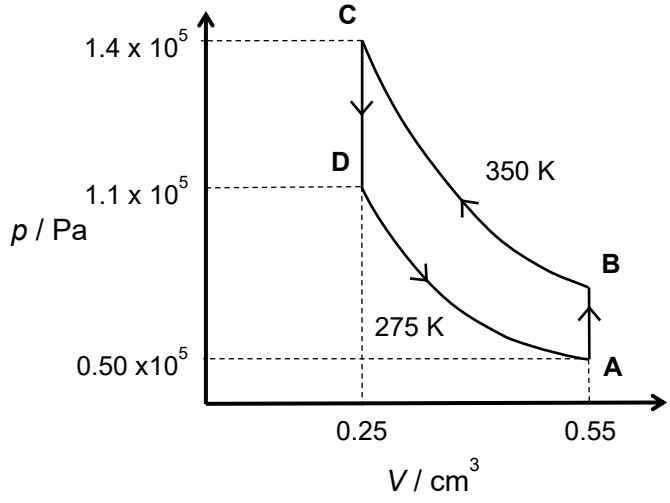

Fig. 3.1

A→B: the gas absorbs heat from the regenerator and its pressure increases.

B→C: the gas contracts isothermally at 350 K

C→D: the gas supplies heat to the regenerator and its pressure decreases

D→A: the gas expands isothermally at 275 K

For Examiner's Use

(i) Determine the pressure at B.

pressure at B = ... Pa [2]

(ii) Show that the change in internal energy from C → D is -11.2 x 10^{-3} J.

[1]

(iii) Fig. 3.2 shows the energy changes for the transitions from A → B → C → D → A.

Complete the table. Show your working clearly.

	Heat supplied / x 10^{-3} J	work done on the gas / x 10^{-3} J	change in internal energy / x 10^{-3} J
A → B			
B → C		27.5	0
C → D			- 11.2
D → A		- 21.6	0

Fig 3.2

[3]

For Examiner's Use

(iv) The above cycle can be used to draw heat out from the surroundings by doing work on the gas. The coefficient of performance (COP) is a measure of how efficient this process is. The COP for such a system is defined as

$$\frac{Q_{in}}{W_{net}}$$

where Q_{in} is the heat energy absorbed by the gas from the surroundings and W_{net} is the **net** work done on the system for one cycle.

1. State which isothermal transition the gas absorbs heat energy from the surroundings.

..... → ..... [1]

2. Hence, determine the COP for this cycle.

COP = ... [2]